

Vivek Jadhav

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SUMMARY

AI, ML & Robotics enthusiast with hands-on experience in **Python**, **Excel**, **Power BI**, and **data analytics**. Skilled in building AI-driven solutions using **machine learning** and **computer vision**, with a focus on real-world applications and insightful dashboards for effective decision-making.

EDUCATION

Symbiosis Institute Of Technology Pune

2022-2026

Bachelor of Technology Robotics & Automation

CGPA : 8.3

EXPERIENCE

Research Intern

July 2024 – April 2025

Symbiosis Centre for Applied Artificial Intelligence (SCAAI)

Pune, India

- Developed a predictive maintenance solution for vertical milling machines using multi-modal machine learning, integrating sensor data to improve accuracy and machine lifespan
- Achieved 97% prediction accuracy and an F1-score of 96%, increasing tool lifespan by 20% through proactive maintenance.

Social Volunteer

JAN 2024 – MAY 2024

Being Volunteer Organization (NGO)

Location

- Coordinated and managed social activities and community outreach events.
- Assisted in project management tasks, including planning and organization
- Activities : Wall-Beautification, Educational sessions for underprivileged children, Feed The Needy.

PROJECTS

Intelligent Facial Recognition for Smart Attendance System | *OpenCV, RasperryPi 5, dlib, Streamlit*

- Integrated Multi-Face Detection and Recognition using Opencv frameworks to enable real-time, contactless attendance tracking, created custom dataset of 85 students
- Deployed a Streamlit app to visualize attendance records and analytics with clear graphs.
- Outperformed Apple's facial recognition on identical twins through optimized embedding generation and detection.

Hybrid Multi-Modal Neural Architecture for Enhanced Wind Turbine Power Forecasting | *Tensorflow*

- Developed a multi-modal deep learning framework combining CNN-LSTM for sensor data and a dense autoencoder for theoretical inputs to forecast wind turbine power output.
- Engineered cyclical wind direction encodings and lag features; achieved optimized results through hyperparameter tuning (R^2 : 0.95, MAE: 180).

Explainable AI: Scene Classification and Grad-CAM Visualization | *ResNet-18, Grad-CAM*

- Built a ResNet-18 model to classify scenes (e.g., forests, glaciers, streets) with 86% accuracy and strong generalization.
- Used Grad-CAM to visualize decision-making, improving model transparency for applications like autonomous navigation.

Motion Capture for Autonomous Swarm Drones | *IR-Checkpoint Localization, Swarm Coordination*

- Developed an IR-based checkpoint system and onboard vision pipeline (Raspberry Pi + OpenCV) for sub-10 cm localization, enabling precise multi-drone waypoint navigation and automatic return-to-checkpoint on disturbance.
- Integrated Pixhawk autopilot (MAVLink) with real-time ground-station data fusion, achieving synchronized swarm flights for surveillance and search-and-rescue on a sub-150 k budget

TECHNICAL SKILLS

Languages: Python, C++,Java(Basic),Embedded C

Frameworks: Langchain,RAG

Developer Tools: Git, Docker, Power BI, Tableau, VS Code, Visual Studio, PyCharm

Machine Learning & A.I.: TensorFlow, Transfer Learning, Natural Language Processing (NLP), Deep Learning, Computer Vision, Data Preprocessing,Machine Learning

Automation Control : Programming Logic Controllers(PLC), Industrial Protocols,Arduino,Cisco Packet Tracer

Project Coordination: Collaboration,Project Management,Problem Solving,Communication

PUBLICATIONS

- **Enhancing Tool Wear Classification Using SMOTE with Ensemble and Deep Learning Techniques**
MethodsX Journal — Paper ID: MEX-D-25-00448 [Under Review]
- **Design and Implementation of a Scalable Network Topology for Smart Megamall**
7th International Conference of Energy, Power and Environment — Paper ID: 903
- **An Approach for High-Speed Network Infrastructure Design for Health Services**
7th International Conference of Energy, Power and Environment
- **Multi-Modal Neural Architecture for Enhanced Wind Turbine Power Forecasting**
1st International Conference on Power and Intelligent Control System
- **Traffic Sign Detection and Recognition for Autonomous Vehicle Navigation by using Satellite Data and Deep Learning Techniques**

CERTIFICATIONS

Decentralized Finance (DeFi) Infrastructure

Duke University

Skills: Blockchain, FinTech, Algorithms

Microsoft Azure SQL

Microsoft

Credential ID: 12GUHYDGYAN6

The Complete LangChain & LLMs Guide

Coursera

Credential ID: CDART3Y2XC3X, Skills: LangChain Integration

GenAI for Customer Communication

Coursera

Credential ID: Y28KW8KAGS9Z, Skills: Generative AI

TensorFlow 2.0 Deep Learning and AI

Udemy

Credential ID: 0004, Skills: TensorFlow, Deep Learning, AI Implementation

BLOGS

Retrieval-Augmented-Generation-framework-with-Langchain-for-chat-with-Docs

www.Rag_hat.com/blog-post-link